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| **PI Name:** | Vora, Tiffany J Ph.D. | | |
| **Project Title:** | Mapping in vivo Protein- DNA Interactions to Unravel Transcription Regulatory Networks | | |
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| **PI Email:** | tvora@Princeton.EDU | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 609-258-5738 |
| **Organization Name:** | Princeton University | | |
| **PI Address 1:** | Princeton University | | |
| **PI Address 2:** | 206 Carl Icahn Laboratory | | |
| **PI Web Page:** | | | |
| **City:** | Princeton | **State:** | NJ |
| **Zip Code:** | 08544 | **Congressional District:** | 12 |
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| **Task Description:** | Transcriptional regulatory networks can be modeled as classical circuit logic diagrams showing the flow of information into and out of the system. I have developed a new technology to monitor this “input,” protein- DNA interactions, on a genome-wide scale in Escherichia coli. The set of cellular DNA-protein interactions under various environmental conditions is visualized by in vivo cross-linking followed by degradation of unbound DNA; protected DNA is queried by hybridization to genomic E. coli microarray and most recently, E. coli genomic tiling arrays. The data gathered in this way will be used to generate “rules” for the construction of synthetic circuits and networks.<br><br>In the future we plan to use our technology to monitor the binding and expression profiles of living cells across a variety of growth, environmental, and genetic conditions. In particular, we anticipate using our technology to unravel the circuit logic underlying the UV-induced DNA-damage response. The logic function of the circuit can be deduced by monitoring the input (DNA-protein interactions) and the output (transcription) of the system. | | |

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| | Studies of this type will elucidate the architectural underpinnings of the cellular transcription network and will promote more accurate in silico models of cellular behavior. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The realization of NASA's vision of mankind's future in the universe requires a commitment to deciphering the most fundamental elements of terrestrial biology. The space environment presents unique challenges to organisms residing therein. Whole-genome technologies such as microarray analysis currently are emerging into space biology research. The technology in this proposal represents a powerful tool preliminary to constructing accurate in silico models of molecular events resulting in appropriate response to a cell's environment. Environments investigated in the future using our technique could include low-gravity, radiation-rich space. A paradigm shift has taken place in recent years. Many researchers now believe that it is possible to accurately model a cell as a system of interconnected networks that obey (after a fashion) the logic rules that are the foundation of other networks such as classic computational and engineering circuit. Thus, the connection between a transcription factor binding to its cis-regulatory sequence in response to an upstream signal and the subsequent transcriptional response can be viewed as a simple circuit where the "input" of the system is the DNA-protein interaction and the "output" is transcription. The logic function of the circuit can be deduced by monitoring the input and output; in the case of transcription factor binding and mRNA production, this mapping is termed "genomic cis-regulatory logic." While this type of analysis has been employed, in a limited fashion to single genes, in systems such as sea urchin development and the yeast cell cycle, it has yet to be applied rigorously and comprehensively to E. coli and other prokaryotic systems. Studies of this type will elucidate the architectural underpinnings of the cellular transcriptional network and will promote more accurate in silico models of cellular behavior. We anticipate using our whole-genome technology to understand the rules of natural transcription regulatory networks. From the knowledge contained in this "rulebook", we further envision the design and construction of synthetic regulatory circuits and networks with diverse functions, ranging from molecular signal detection to environmental engineering to technological advances yet unimagined. These technologies would impact life and research, on Earth and in space, for the better. |
| **Task Progress:** | Monitoring in vivo Protein-DNA Interactions to Unravel Transcription Regulatory Networks<br>This project involves the de novo development of a novel genomic technology designed to generate a whole-genome set of protein-DNA "footprints" similar to those produced by chromatin immunoprecipitation (ChIP). Location analyses via ChIP are characterized by the use of antibodies specific to the protein under study. However, this technique cannot accomplish a mapping of the binding sites of the complete repertoire of DNA-binding proteins.<br><br>We have developed a technology to monitor simultaneously the in vivo occupancy of E. coli's genomic set of binding sites. ChIP followed by microarray analysis combines the direct detection of DNA-protein complexes of interest with the statistical power of whole-genome quantitation by microarray hybridization. Furthermore, we have refined the experimental methodology and computational tools such that the theoretical resolution of the system is 50 base pairs or less; thus this technology has the capability to provide the most detailed binding information of any genome-scale study to date.<br><br>Formaldehyde, a common cross-linking agent, is used to produce protein-protein and DNA-protein crosslinks at a resolution of approximately two angstroms. Once crosslinking is complete, a sonicator is used to simultaneously lyse the cells and fragment the DNA to an average size of one kilobase. The pool of DNA and DNA-protein complexes is then exposed to DNaseI, destroying the strands of DNA that are not sterically protected from digestion. To obtain a pool of DNA further enriched for DNA-protein complexes over unbound DNA, a modified phenol-chloroform extraction is carried out. A protein-DNA complex is more hydrophobic than free DNA alone; as a result, free DNA moves to the aqueous phase during extraction, while DNA-protein complexes migrate to the interface of the aqueous and organic phases. Retention of the interface results in the collection of DNA-protein complexes that existed in vivo at the time of crosslinking. Finally, the crosslinks are reversed by heat, and the DNA is purified, labeled, and hybridized to an Affymetrix array to provide a quantitative measurement of the in vivo protein occupancy of a given DNA fragment.<br><br>We used multiple genome-wide analyses to discover patterns in the protein occupancy across the genome. We found that the genes that displayed similar protein occupancy patterns by several different metrics – including median open reading frame (ORF) binding, and binding around the ATG – were enriched in functional categories, especially that of tRNA. When we added in data from other sources, such as RNAP occupancy and gene expression, we found other complex patterns emerging. To our surprise, we found sets of genes that showed significant protein occupancy downstream of the ATG. Only a very small set of genes are known to be regulated downstream in E. coli, and as yet we have not ascertained the common element among these additional genes.<br><br>The initial goal of this project was to develop a technology that would monitor the occupancy of single cis-regulatory elements. In a rough sense, we hoped to assign a continuous measure of occupancy, in a population, to each of these elements. Local analyses of protein occupancy would provide a genomic map of "inputs" to transcription regulatory networks. We have pushed our in vivo genomic footprinting technology to a DNA fragment size of less than 100 base pairs on average, far below any published resolution. Nonetheless, we have discovered that the local occupancy of genomic sequences is far more complex than previously realized. We have identified well-defined instances of DNA-protein interactions that dynamically change across growth conditions, metabolic conditions, and genotypes. We are in the process of joining these regulatory "inputs" to the tiling-resolution RNA abundance data we have recently acquired using the same Affymetrix microarray. By joining the "inputs" and "outputs" of genetic regulatory networks, we are in the process of modeling the complex series of decisions carried out by the cell.<br><br>Our in vivo protein occupancy display methodology also allowed us to gain insight into one of the outstanding questions of microbial biology: how bacterial cells, specifically Escherichia coli, compact a 4.6 Mb genome into a cell only 2 – 3 microns in length, and yet still maintain the DNA in a form that is accessible to the transcription, translation, repair, and replication machineries. Furthermore, the genome must be organized such that a single DNA lesion, such as a nick, does not lead to total loss of viability. We have integrated our in vivo protein occupancy data with genome-wide measures of RNA abundance, biophysical properties, evolutionary conservation, known and predicted protein binding sites, and accessibility of DNA. We believe that we have discovered "bacterial heterochromatin", regions of DNA that are tightly packed with protein and maintain the genome in an ordered structure. The development of this novel model of genome organization is a direct result of our ability to monitor the entire genomic repertoire of DNA-protein |

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| | interactions in live cells. Overall, this research project has been focused on creating a new, unbiased, whole-genome set of experimental and computational techniques intended to investigate the genomic repertoire of DNA-protein interactions in vivo. We have utilized the systems biology paradigm of multiple systems integration to create a new picture of the processes underlying the simplicity and elegance of the biology of E. coli. By combining many types of data on a genome-wide scale, we have established a promising research platform for unraveling the elegant genetic networks underlying the complex physiology of microbial cells. |
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